PRIMARY CUTANEOUS T-CELL LYMPHOMAS

Cutaneous T-cell lymphomas (CTCLs) are non-Hodgkin lymphomas characterized by a dominant skin-homing T-cell clone. They account for approximately 80% of all cutaneous lymphomas. The most common subtypes are mycosis fungoides (MF) and Sézary syndrome (SS), with an annual incidence of three to four new cases per million people. After MF and SS, the primary cutaneous CD30+ lymphoproliferative disorders—including lymphomatoid papulosis (LyP) and primary cutaneous anaplastic large cell lymphoma (cALCL)—represent the second most common group of CTCLs, comprising approximately 30% of cases.

An understanding of the unique epidermal sanctuary that harbors, preserves, and perpetuates CTCL is essential for comprehending disease persistence and recurrence.

Mycosis Fungoides

DEFINITION

Mycosis fungoides (MF) is the most frequent variant of CTCL, typically presenting in mid to late adulthood (median age at diagnosis, 55 to 60 years), with a male-to-female ratio of 2:1.

CLINICAL MANIFESTATIONS: Skin Signs

MF is clinically categorized into patch, plaque, and tumor stages, although patients may exhibit more than one type of lesion simultaneously.

Patch Stage

In early patch-stage MF, patients present with single or multiple erythematous, scaly macules and patches of variable size, usually well-demarcated. Lesion color ranges from orange to dusky violet-red. The distribution typically favors non-sun-exposed areas, with early involvement of the "bathing trunk" region and intertriginous zones. The eruption may be intensely pruritic or asymptomatic and can occasionally be transient, resolving spontaneously without scarring.

Diagnosis at this stage is often challenging. Many patients report a history of "chronic dermatitis" lasting 10 to 20 years, previously misdiagnosed as contact dermatitis, atopic dermatitis, psoriasis, or eczema. In any case of therapy-resistant dermatosis, multiple skin biopsies should be performed to evaluate for MF.

Plaque Stage

Patches may persist for months or years before progressing to plaques, or plaques may arise de novo. Plaques are sharply demarcated, scaly, elevated lesions that are dusky red to violaceous and variably indurated. Annular, arcuate, or serpiginous configurations with central atrophy and active disease at the periphery are characteristic. Central clearing may occur, leaving poikiloderma—comprising atrophy, telangiectasia, and pigmentary changes (hyper- or hypopigmentation).

Tumor Stage

Tumors may develop anywhere but show a predilection for the face and body folds—such as the axillae, groin, antecubital fossae, and, in women, the inframammary area. Tumors may arise within pre-existing patches or plaques, indicating a vertical growth phase. This transition reflects more biologically aggressive behavior, with neoplastic cells expanding vertically through the dermis, leading to rapidly enlarging nodules or tumors.